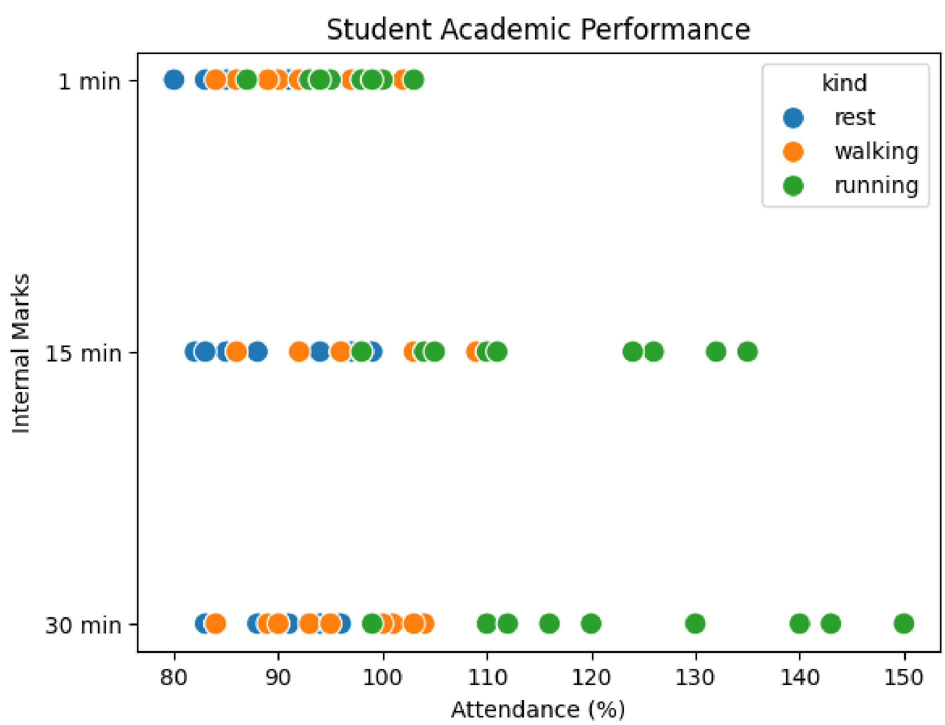
Data Visualization Lab – Activity 1

Scenario 1 – Student Academic Performance Dashboard

Code

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

url = "https://raw.githubusercontent.com/mwaskom/seaborn-
data/master/exercise.csv"
df = pd.read_csv(url)

sns.scatterplot(data=df,
                x="pulse",
                y="time",
                hue="kind",
                s=100)

plt.title("Student Academic Performance")
plt.xlabel("Attendance (%)")
plt.ylabel("Internal Marks")
plt.show()
```

Output

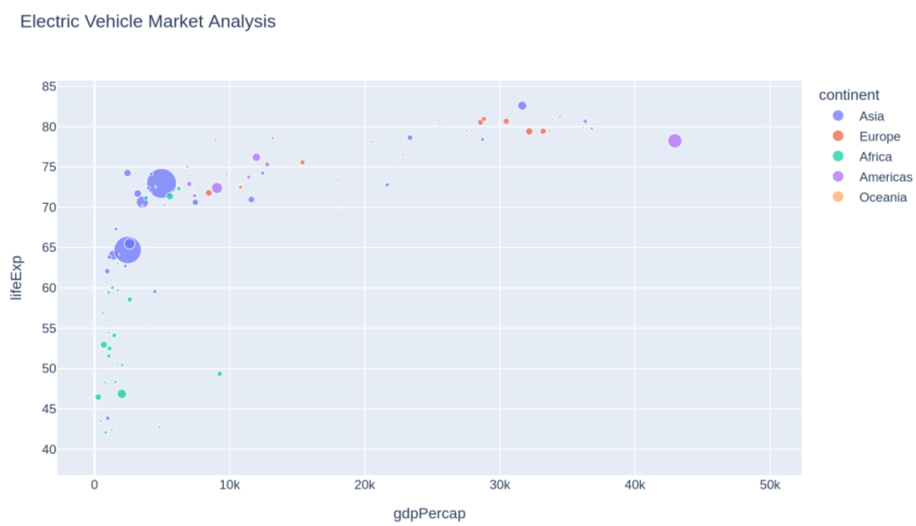


Scenario 2 – Electric Vehicle Market Analysis

Code

```
import plotly.express as px

df = px.data.gapminder().query("year==2007")

fig = px.scatter(df,
                 x="gdpPercap",
                 y="lifeExp",
                 size="pop",
                 color="continent",
                 title="Electric Vehicle Market Analysis")

fig.show()
```

Output

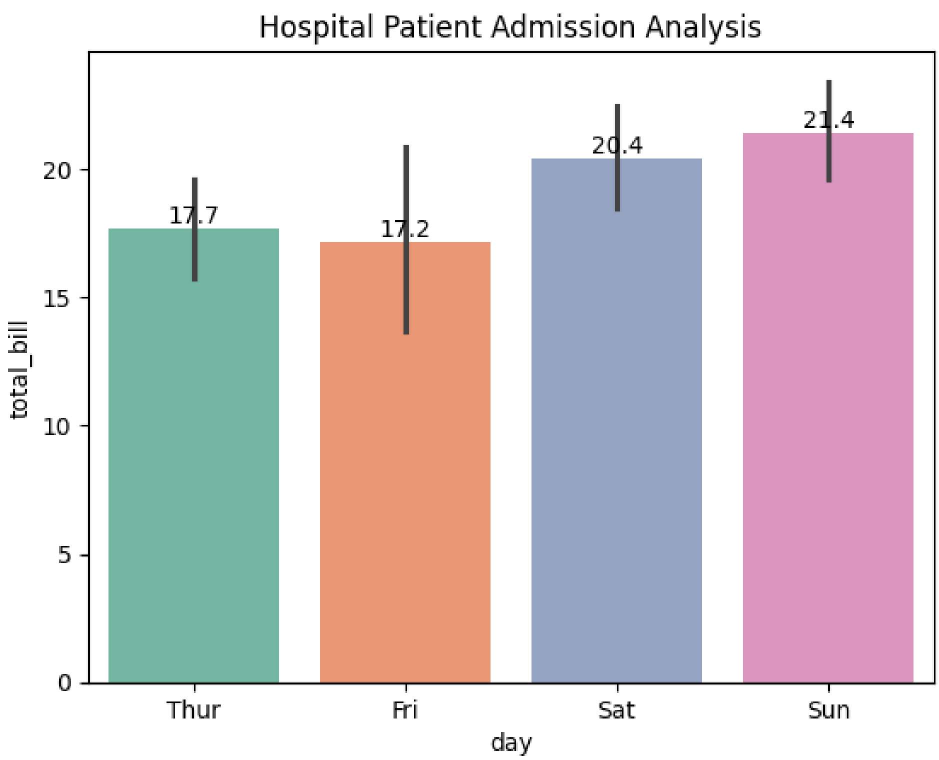


Scenario 3 – Hospital Patient Admission Analysis

Code

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")

ax = sns.barplot(data=tips,
                 x="day",
                 y="total_bill",
                 hue="day",
                 palette="Set2")

for container in ax.containers:
    ax.bar_label(container, fmt="%.1f")

plt.title("Hospital Patient Admission Analysis")
plt.show()
```

Output

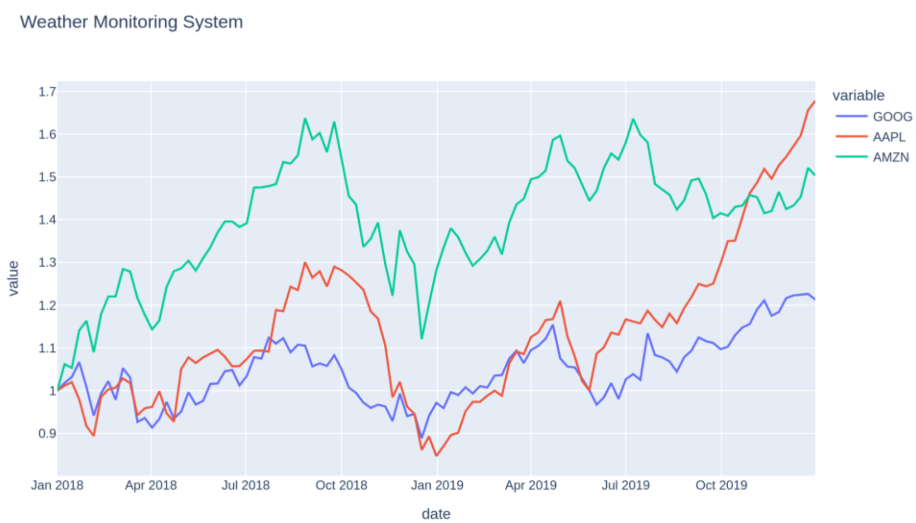


Scenario 4 – Weather Monitoring System

Code

```
import plotly.express as px

df = px.data.stocks()

fig = px.line(df,
              x="date",
              y=["GOOG","AAPL","AMZN"],
              title="Weather Monitoring System")

fig.show()
```

Output

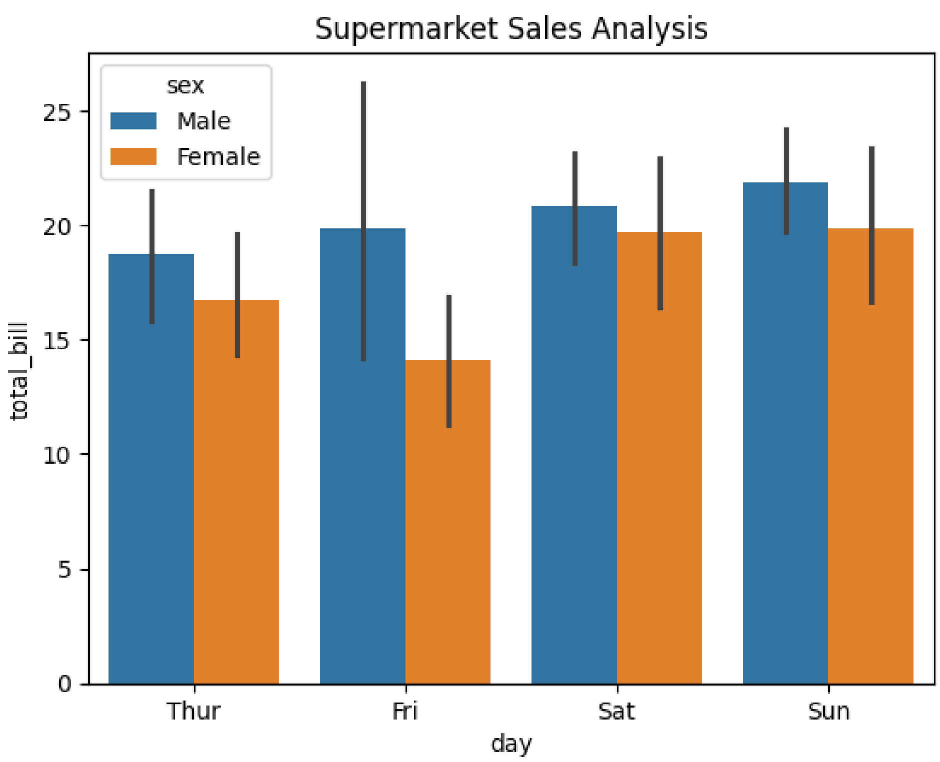


Scenario 5 – Supermarket Sales Analysis

Code

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")

sns.barplot(data=tips,
            x="day",
            y="total_bill",
            hue="sex")

plt.title("Supermarket Sales Analysis")
plt.show()
```

Output

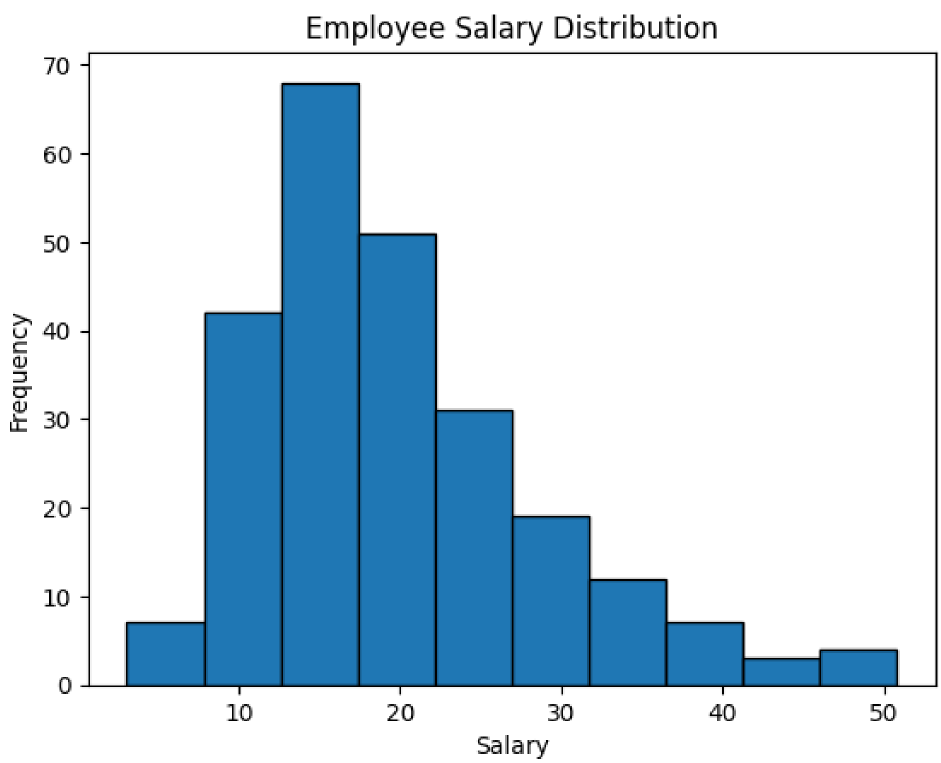


Scenario 6 – Employee Salary Distribution

Code

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")

plt.hist(tips["total_bill"],
         bins=10,
         edgecolor="black")

plt.title("Employee Salary Distribution")
plt.xlabel("Salary")
plt.ylabel("Frequency")
plt.show()
```

Output

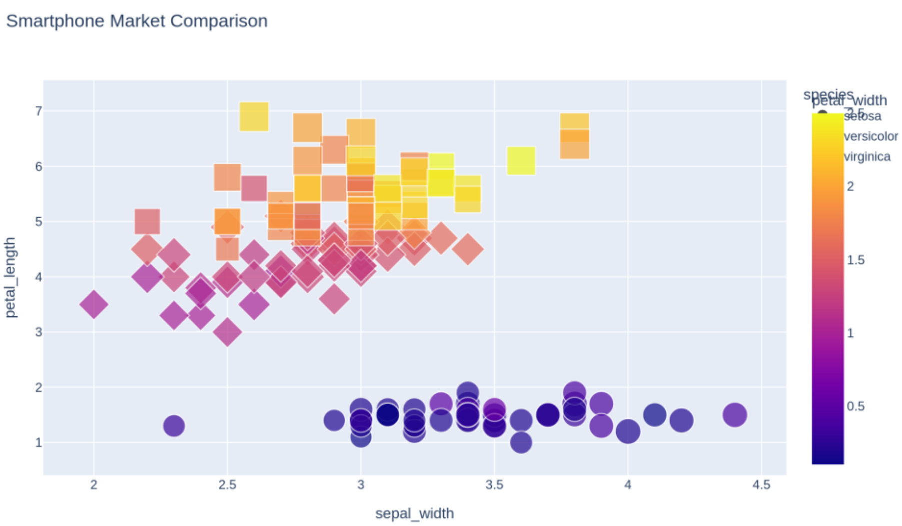


Scenario 7 – Smartphone Market Comparison

Code

```
import plotly.express as px

df = px.data.iris()

fig = px.scatter(df,
                 x="sepal_width",
                 y="petal_length",
                 color="petal_width",
                 symbol="species",
                 size="sepal_length",
                 title="Smartphone Market Comparison")

fig.show()
```

Output

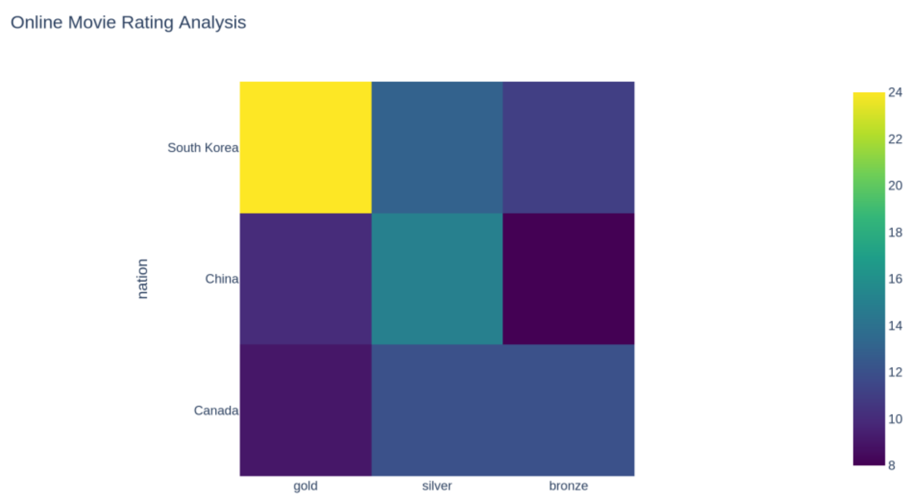


Scenario 8 – Online Movie Rating Analysis

Code

```
import plotly.express as px

df = px.data.medals_wide()

fig = px.imshow(df.set_index("nation"),
                color_continuous_scale="Viridis",
                title="Online Movie Rating Analysis")

fig.show()
```

Output

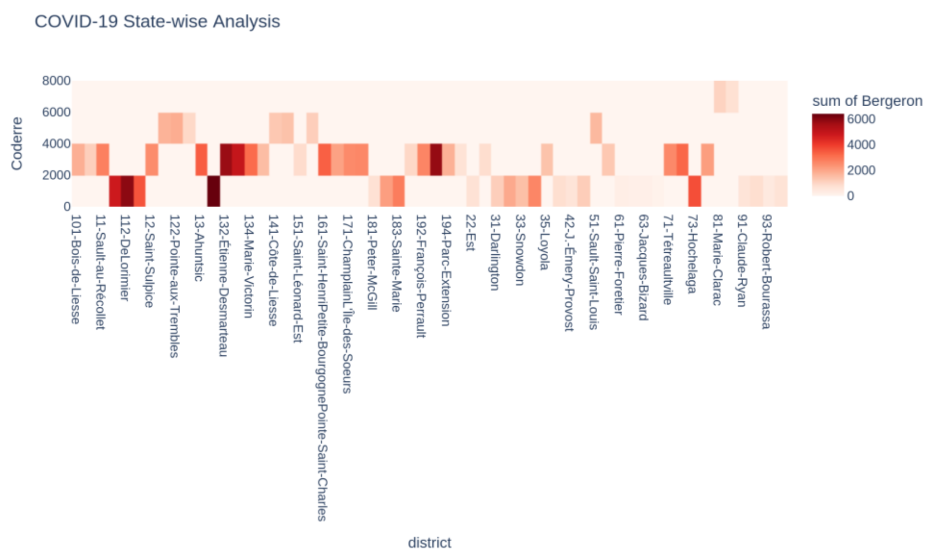


Scenario 9 – COVID-19 State-wise Analysis

Code

```
import plotly.express as px

df = px.data.election()

fig = px.density_heatmap(df,
                         x="district",
                         y="Coderre",
                         z="Bergeron",
                         color_continuous_scale="Reds",
                         title="COVID-19 State-wise Analysis")

fig.show()
```

Output



Scenario 10 – Iris Flower Classification

Code

```
import plotly.express as px

df = px.data.iris()

fig = px.scatter(df,
                 x="sepal_length",
                 y="petal_length",
                 color="species",
                 symbol="species",
                 size="sepal_width",
                 title="Iris Flower Classification")

fig.show()
```

Output